

1) Write the Java code to divide the number in to half without using /

n = 15
answer = 7

right shift (>>) and left shift operator(<<)

n = 15(decimal no)

15 ==> 1111
8 4 2 1

15 1111 >> 1
7 0111

11 ==> 1011
1011 > 1

5 0101

15 8 4 2 1
1 1 1 1 > 1

15 ==> 1 1 1 1 >> 1
8 4 2 1

0 1 1 1
3 2 1 0

n = 10

2) Write the Java code to double the number without using *

3) write the java code to count the total no of set bits in a number

set bit ==> 1

n = 15 ==> 1 1 1 1

n = 13 ==> 1 1 0 1

n = 13

1 ==> 0001

1 1 0 1 & 0 0 0 1

1 1 0 1
0 0 0 1
1 -> 3

& -> 00
& -> 10

1 1 0 1 >> 1

0 1 1 0 0 0 1 1 0 0 0 1 0 0 0 0
0 0 0 1 0 0 0 1 0 0 0 1 0 0 0 1

HashMap gives the output not in the insertion order, it gives the output in bucket order

HashMap will internally use HashTable to store the key value pairs

map.put(50, "A");
map.put(10, "B");
map.put(40, "C");
map.put(20, "D");

50 & 15

0	
1	
2	{ 50, "A" }
3	
4	{ 20, "D" }
5	
6	
7	
8	{ 40, "C" }
9	
10	{ 10, "B" }
11	
12	
13	
14	
15	

HashTable

construct a linked hash map to store the department name as key and correspondng HOD faculty ad the value, and print them in same order of insertion

Raju ==> string

B ==> character

'B' ==> Characater or "B" or "Raju" ==> String

TreeMap ==> handle sorting logic

by default, treemap will work in asceding order only